

The family of maximal minors in increasing row order

MathlibAnnex.Matrix.maximalMinors

def

Collects the determinants of all square submatrices with all columns and an increasing selection of rows.

Statement

Let R be a commutative ring, I a linearly ordered set, and A an I -by- n matrix over R . The family of maximal minors is

$$\begin{aligned} \mathcal{M}(A) : \{S \subseteq I : |S| = n\} &\longrightarrow R, \\ S = \{s_1 < \cdots < s_n\} &\longmapsto \det((A_{s_i, j})_{1 \leq i, j \leq n}). \end{aligned}$$

Thus the rows are taken in the indicated increasing order, and the columns remain in their original order. Only finite subsets of size n occur in the domain.

Definition

For every selected subset, the ordering convention is part of the defining formula:

$$\begin{aligned} S = \{s_1 < \cdots < s_n\}, \quad A_S &:= (A_{s_i, j})_{1 \leq i, j \leq n}, \\ \mathcal{M}(A)(S) &:= \det A_S. \end{aligned}$$

The increasing enumeration fixes the row order and hence the sign. This is not the determinant of rows arranged in an unspecified order.

Assumptions

The number n is any nonnegative integer. The row set I is linearly ordered and may be infinite. No field, nonzero-minor or rank assumption is made.

Conclusion

Each component is a determinant with the same increasing-row convention. When $n = 0$, the unique selected subset is empty and its empty determinant is 1. If there are no n -element subsets, the family has empty domain.

An increasing injection from the n column positions into I determines its range subset; the cited adapter lemmas identify the selected matrix and minor with that injection. An arbitrary ordered tuple of rows carries additional order information and is treated separately in the factorization theorem. Selection also commutes with right multiplication: $(AL)_S = A_S L$.

Main citations

- Exact declaration and its source — `MathlibAnnex.Matrix.maximalMinors`
- Subsets indexing maximal minors — `MathlibAnnex.Matrix.MaximalMinorIndex`
- Increasing enumeration of selected rows — `MathlibAnnex.Matrix.MaximalMinorIndex.orderedRows`
- An increasing injection supplies an index —
`MathlibAnnex.Matrix.MaximalMinorIndex.ofOrderEmbedding`
- The enumeration agrees with the increasing injection —
`MathlibAnnex.Matrix.MaximalMinorIndex.orderedRows_ofOrderEmbedding`
- The selected square submatrix — `MathlibAnnex.Matrix.maximalSubmatrix`
- The determinant of the selected submatrix — `MathlibAnnex.Matrix.maximalMinor`
- Submatrix selection through an increasing injection —
`MathlibAnnex.Matrix.maximalSubmatrix_ofOrderEmbedding`
- Minor selection through an increasing injection —
`MathlibAnnex.Matrix.maximalMinor_ofOrderEmbedding`
- Selection commutes with right multiplication — `MathlibAnnex.Matrix.maximalSubmatrix_mul`

Lean source signature (exact)

```
def maximalMinors {n : ℕ} {ι : Type u} {R : Type v} [LinearOrder ι] [CommRing R]
  (A : _root_.Matrix ι (Fin n) R) : MaximalMinorIndex n ι → R :=
  fun s => maximalMinor A s
```

In the source	Mathematical meaning
<code>{n : ℕ} {ι : Type u} [LinearOrder ι] {R : Type v} [CommRing R]</code>	The nonnegative integer n , linearly ordered row set I and commutative coefficient ring R . The whole row set may be infinite.
<code>(A : root.Matrix ι (Fin n) R)</code>	The I -by- n matrix A , whose columns keep their given order.
<code>MaximalMinorIndex n ι</code>	The finite row subsets $S \subseteq I$ with $ S = n$, as defined in the cited index declaration.

In the source	Mathematical meaning
<code>MaximalMinorIndex n ι → R</code>	The output family $\mathcal{M}(A)$, assigning one scalar to each such subset; it is not a single determinant.
<code>fun s => maximalMinor A s</code>	The entire RHS: at $S = \{s_1 < \dots < s_n\}$ return $\det A_S$, where $A_S = (A_{s_i, j})$. The cited <code>orderedRows</code> and <code>maximalSubmatrix</code> use this increasing order. At $n = 0$ the empty determinant is 1.

▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section
```

```
open scoped BigOperators
```

```
namespace MathlibAnnex
```

```
namespace Matrix
```

```
universe u v
```

▼ Exact content identity

Declaration: `MathlibAnnex.Matrix.maximalMinors`

Accepted content SHA-256: `e36602fe7b0cea677f7e40dd18e13681e6a283ead24981066f7dba1e68b10ec7`

Accepted source guide SHA-256: `8d0a2a88b83a5a141ec526ab489ab034737bc88227364e7f3262cf2946c560e8`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`